David Erik Cunningham, Ph.D., P.Eng.

davidcunningham1998@gmail.com | 1-226-235-4860

https://www.linkedin.com/in/davidcunningham-phd

Professional Experience

Electro-Mechanical Equipment Research & Development Engineer

Canadian Nuclear Laboratories – Chalk River, ON | Oct 2024 – Present

- Led and coordinated cross-functional engineering and operations teams, driving strategic development and scale-up of Actinium-225 isotope manufacturing.
- Designed custom tooling using ANSYS simulations to optimize safety-critical radiochemical processes and reduce operational risks.
- Planned and executed experimental test programs integrating radiological safety, process control, and statistical quality methods.
- Implemented active source test loops with health physics teams, reducing radiological exposure and improving safety margins.
- Applied ISO 9001, CSA N299, BTH-1, and ASME Section VIII standards to ensure compliance and system reliability.
- Advanced nuclear medicine supply chains by developing novel methodologies for isotope production.
- Conducted investigative testing of advanced heat exchangers for molten salt reactors, qualifying AM components for next-gen nuclear use.
- Performed SEM microstructural analysis, dimensional inspections, and mechanical testing of prototypes.
- Authored R&D proposals, business cases, and cost estimates; presented strategic recommendations and technical findings to executive leadership and regulatory bodies to inform decision-making.
- Specialized expertise: FEA of impact/drop scenarios, Ogden-model hyperelastic foams, irradiation/failure analyses, and high-reliability nuclear equipment.

Orthopaedic Implant Research & Development Engineer

Roth McFarlane Hand and Upper Limb Clinic – London, ON | Sep 2021 – Oct 2024

- Developed finite element models to evaluate implant–bone interactions under physiological loads.
- Created custom Python programs to automate the design and analysis pipeline, including:
    - Automated generation of varying stemless Reverse Total Shoulder Arthroplasty (RTSA) designs Automated FE joint model setup (geometry creation, implant positioning, partitioning, meshing, and simulation execution)

    - Advanced post-processing workflows for comprehensive evaluation of bone anatomy behavior and implant performance
- Managed and mentored multi-disciplinary R&D teams, overseeing project planning, resource allocation, and integration of data-driven approaches to advance implant design initiatives.
- Designed experimental loading systems and integrated machine vision software for micromotion/kinematic testing.
- Increased throughput via custom high performance computing builds for simulation and data processing.
- Authored, peer-reviewed papers, reports, and conference presentations advancing implant design science.
- Built 3D visualization and automation tools (Abaqus API, VAEX, Plotly, Pandas) for large-scale dataset analysis.

Mechanical Reliability Engineering EIT

NOVA Chemicals – Corunna, ON | Jan 2019 – Aug 2019

- Implemented and managed condition monitoring strategies, increasing asset reliability and operational uptime across critical systems.
- Contributed to IoT integration of remote condition monitoring devices in high-risk locations on-site.
- Conducted vibration analyses (AMS Machinery Health), enabling predictive maintenance.
- Developed and executed preventative maintenance response plans.
- Integrated Emerson Triaxial Accelerometers, improving diagnostics, operational efficiency, and condition monitoring reliability across critical assets.
- Directed cross-departmental teams in executing modifications on high-pressure rotating equipment, ensuring continuous production while enhancing safety and performance.
- Performed gas/radiation testing per OSHA, API, and NOVA standards.
- Performed safety audits on high pressure and caustic rotation equipment to ensure safety compliance.
- Troubleshot and repaired rotating machinery, restoring functionality under critical timelines.

Polyethylene & Capital Engineering EIT

NOVA Chemicals – Mooretown, ON | May 2018 – Dec 2018

- Designed ASME-compliant pressure vessels and piping systems.
- Conducted stress analyses (CAEPIPE, CAESAR II, PVElite) for optimized designs in piping systems.
- Evaluated vibration assessments for start-up and critical shutdown scenarios.
- Planned research and design of safety-critical devices (heat exchangers, dampeners, filters).
- Performed low-temperature embrittlement assessments and issued construction packages for capital project enabling.

- Coordinated contractors and multidisciplinary engineering teams, ensuring successful delivery of capital projects on time, within budget, and aligned with organizational objectives.

Manufacturing Engineering Systems Designer

Schneider Electric – Victoria, BC | May 2016 – Aug 2017

- Partnered with senior leadership to identify and resolve operational barriers within the pilot meter program, improving organizational efficiency.
- Developed and deployed pilot unit scheduling software, streamlining program management and supporting continuous improvement initiatives
- Reviewed and analyzed operating procedures across a Six Sigma–driven plant environment and recommended Lean Six Sigma improvements.
- Operated high-voltage testing equipment (3300V+, LabView-controlled) to perform rigorous quality control and data analysis.
- Applied Lean tools (5S, Kanban, Poka-yoke, Jidoka) to streamline assembly/testing workflows.
- Operated conformal coating machinery for the heat treatment of environmentally sensitive circuitry.

Certifications

- Professional Engineer (P.Eng.), Ontario – License #100630934
- Lean Six Sigma Green Belt – CN: 40806 (2024)
- Lean Six Sigma Yellow Belt – CN: 40432 (2024)
- SOLIDWORKS CAD Associate (CSWA) – 2023
- SOLIDWORKS Simulation Associate (CSWA-S) – 2020

Publications

- An Analysis of Primary Stability of Stemless Humeral Implants in Shoulder Arthroplasty, *Western Electronic Thesis and Dissertation Repository*, https://ir.lib.uwo.ca/etd/10095
- Stemless Reverse Shoulder Arthroplasty Neck-Shaft Angle Influences Humeral Component Time-Zero Fixation and Survivorship: A Cadaveric Biomechanical Assessment, *Journal of Shoulder and Elbow Surgery - International*, https://doi.org/10.1016/j.jseint.2024.04.001
- Stemless Reverse Humeral Component Neck Shaft Angle has an Influence on Initial Fixation, *Journal of Shoulder and Elbow Surgery*, https://doi.org/10.1016/j.jse.2023.06.035
- Leveraging Kindness in Canadian Post-Secondary Education: A Conceptual Paper, *College Teaching*, https://doi.org/10.1080/87567555.2023.2181307

Technical Skills

Leadership and Management Skills

- Strategic project planning & execution, cross-functional team leadership, budgeting & resource allocation, regulatory & stakeholder engagement, continuous improvement & Lean Six Sigma

Engineering & Simulation

- ABAQUS (CAE/API), ANSYS (Mechanical/Thermal,LS-DYNA), SolidWorks (CSWA, CSWA-S), Autodesk Inventor, PVElite, CAESAR II, Mimics, MicroStation

Programming & Automation

- Python, C++, C, Arduino, MATLAB, SQL, VBA, HTML
- Data analysis & visualization: Pandas, Numpy, SciPy, Scikit-Learn, VAEX, Plotly, SPSS

Manufacturing & Reliability

- DMLS/FDM additive design, LabVIEW, SAP, SmartPlant Foundation, AMS Machinery Health, Rapid Prototyping

Design & Quality Tools

- DFMEA, DFA, DFM, QFD, FTA, HAZOP, TRIZ, CPM, Stage Gate, GD&T

Standards & Compliance

- ASME B31.3, ASME Sec III, ASME Sec VIII Div 1, ISO 9001, CSA N299, OSHA, API, ASTM, WHMIS, Six Sigma, Kaizen